

$$\therefore \text{Required percentage} = \left(\frac{10}{50} \times 100 \right) \% = 20\%.$$

28. Let the given number be x .

$$\text{Then, } x^2 - 25 = (x - 25)^2 \Leftrightarrow x^2 - 25 = x^2 + 625 - 50x \Leftrightarrow 50x = 650 \Leftrightarrow x = 13.$$

29. $(11)^2 = 121$ and $(17)^2 = 289$.

So, the perfect squares between 120 and 300 are the squares of numbers from 11 to 17. Clearly, these are 7 in number.

30. The first perfect square number after 50 is 64 ($= 8^2$) and the last perfect square number before 1000 is 961 [$= (31)^2$].

So, the perfect squares between 50 and 1000 are the squares of numbers from 8 to 31. Clearly, these are 24 in number.

31. Clearly, the man was born between 1800 and 1850.

The only perfect square number between 1800 and 1850 is 1849. And, $1849 = (43)^2$.

So, the man was 43 years old in 1849. Thus, he was born in $(1849 - 43) = 1806$.

$$32. Q = \sqrt{\frac{(8R)^2}{4}} = \frac{\sqrt{(8R)^2}}{\sqrt{4}} = \frac{8R}{2} = 4R.$$

33. The smallest such number is 1444 [$= (38)^2$]. It lies between 1000 and 2000.

$$34. \left(\sqrt{2} + \frac{1}{\sqrt{2}} \right)^2 = (\sqrt{2})^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + 2 \times \sqrt{2} \times \frac{1}{\sqrt{2}} \\ = 2 + \frac{1}{2} + 2 = 4\frac{1}{2}.$$

35. We have:

$$1 \times 2 \times 3 \times 4 = 24 \text{ and } 24 + 1 = 25 [= 5^2].$$

$$2 \times 3 \times 4 \times 5 = 120 \text{ and } 120 + 1 = 121 [= 11^2]$$

$$3 \times 4 \times 5 \times 6 = 360 \text{ and } 360 + 1 = 361 [= 19^2]$$

$$4 \times 5 \times 6 \times 7 = 840 \text{ and } 840 + 1 = 841 [= 29^2].$$

$$\therefore p = 1.$$

$$36. \sqrt{0.16} = \sqrt{\frac{16}{100}} = \frac{\sqrt{16}}{\sqrt{100}} = \frac{4}{10} = 0.4.$$

$$37. \sqrt{0.000441} = \sqrt{\frac{441}{10^6}} = \frac{\sqrt{441}}{\sqrt{10^6}} = \frac{21}{10^3} = \frac{21}{1000} = 0.021.$$

$$38. \sqrt{0.00004761} = \sqrt{\frac{4761}{10^8}} = \frac{\sqrt{4761}}{\sqrt{10^8}} = \frac{69}{10^4} = \frac{69}{10000} = 0.0069.$$

$$39. 1.5^2 \times \sqrt{0.0225} = 1.5^2 \times \sqrt{\frac{225}{10000}} = 2.25 \times \frac{15}{100} \\ = 2.25 \times 0.15 = 0.3375.$$

$$40. \sqrt{0.01} + \sqrt{0.0064} = \sqrt{0.01} + \sqrt{\frac{64}{10000}} \\ = \sqrt{0.01} + \frac{8}{100} = \sqrt{0.01 + 0.08} = \sqrt{0.09} = 0.3.$$

$$41. \text{Given exp.} = \sqrt{\frac{1}{100}} + \sqrt{\frac{81}{100}} + \sqrt{\frac{121}{100}} + \sqrt{\frac{9}{10000}} \\ = \frac{1}{10} + \frac{9}{10} + \frac{11}{10} + \frac{3}{100} \\ = 0.1 + 0.9 + 1.1 + 0.03 = 2.13.$$

$$42. \text{Given exp.} = \sqrt{\frac{25}{10000}} \times \sqrt{\frac{225}{100}} \times \sqrt{\frac{1}{10000}} \\ = \frac{5}{100} \times \frac{15}{10} \times \frac{1}{100} = \frac{75}{10000} = 0.00075.$$

$$43. \begin{array}{r|l} 1 & 1.5\overline{6} \ 25 \ (1.25) \\ & 1 \\ \hline 22 & 56 \\ & 44 \\ \hline 245 & 12 \ 25 \\ & 12 \ 25 \\ \hline & \times \end{array} \quad \therefore \sqrt{1.5625} = 1.25.$$

$$44. \sqrt{6760000} = \sqrt{0.00000676 \times 10^{12}} = \sqrt{0.00000676} \times \sqrt{10^{12}} \\ = .0026 \times 10^6 = 2600.$$

$$45. \text{Given exp.} = \sqrt{\frac{18225}{10^2}} + \sqrt{\frac{18225}{10^4}} + \sqrt{\frac{18225}{10^6}} + \sqrt{\frac{18225}{10^8}} \\ = \frac{\sqrt{18225}}{10} + \frac{\sqrt{18225}}{10^2} + \frac{\sqrt{18225}}{10^3} + \frac{\sqrt{18225}}{10^4} \\ = \frac{135}{10} + \frac{135}{100} + \frac{135}{1000} + \frac{135}{10000} \\ = 13.5 + 1.35 + 0.135 + 0.0135 = 14.9985.$$

$$46. \text{Given exp.} = \sqrt{\frac{4096}{10^2}} + \sqrt{\frac{4096}{10^4}} + \sqrt{\frac{4096}{10^6}} + \sqrt{\frac{4096}{10^8}} \\ = \frac{\sqrt{4096}}{10} + \frac{\sqrt{4096}}{10^2} + \frac{\sqrt{4096}}{10^3} + \frac{\sqrt{4096}}{10^4} \\ = \frac{64}{10} + \frac{64}{100} + \frac{64}{1000} + \frac{64}{10000} \\ = 6.4 + 0.64 + 0.064 + 0.0064 \\ = 7.1104 \approx 7.11.$$

$$47. \text{Given exp.} = \sqrt{1.30} + \sqrt{1300} + \sqrt{0.0130} \\ = \sqrt{\frac{130}{100}} + \sqrt{13 \times 100} + \sqrt{\frac{130}{10000}} \\ = \frac{\sqrt{130}}{10} + \sqrt{13} \times 10 + \frac{\sqrt{130}}{100} \\ = \frac{11.40}{10} + 3.605 \times 10 + \frac{11.40}{100} \\ = 1.14 + 36.05 + 0.114 = 37.304.$$

$$48. \frac{52}{x} = \sqrt{\frac{169}{289}} \\ \Leftrightarrow \frac{52}{x} = \frac{13}{17} \\ \Leftrightarrow x = \left(\frac{52 \times 17}{13} \right) = 68.$$

49. Let the missing number be x .

$$\text{Then, } x^2 = 15 \times 135 \Leftrightarrow x = \sqrt{15 \times 135} \\ = \sqrt{15^2 \times 3^2} = 15 \times 3 = 45.$$